

create a `vault.yml` file as indicated below:

```
$ ansible-vault create inventory/kvm/group_vars/all/vault.yml
```

Provide a password when prompted, following which, the Nano text editor will open. You can enter the following credentials and save the file:

```
---
vault_mysql_cacti_password_hash:
  "528573A4E6FE4F3E88455F2F060EB6F63ECBECAA"

vault_mysql_user: "cacti"
vault_mysql_password: "cacti123"
```

You can edit the same file, if you wish, using the following command:

```
$ ansible-vault edit inventory/kvm/group_vars/all/vault.yml
```

It will prompt you for a password, and on successful authentication, your text editor will open with the decrypted file contents for editing.

Apache

Cacti has many dependency packages, and the first software that we will install is the Apache HTTP server.

```
---
- name: Install web server
  hosts: centos
  gather_facts: true
  tags: [httpd]

  tasks:
    - name: Update the software package repository
      yum:
        name: '*'
        update_cache: yes

    - name: Install HTTP packages
      package:
        name: "[{ item }]"
        state: latest
      with_items:
        - wget
        - nano
        - httpd
        - httpd-devel

    - name: Start the httpd server
      service:
        name: httpd
        state: started
```

```
- wait_for:
  port: 80
```

A 'yum update' is first performed to sync with the package repositories. The `httpd` Web server and a few other packages are then installed. The server is started, and the Ansible playbook waits for the server to listen on port 80.

MySQL and PHP

The MySQL, PHP and RRDTool packages are then installed, following which the SNMP and MySQL servers are started as shown below:

```
- name: Install MySQL, PHP packages
  hosts: centos
  become: yes
  become_method: sudo
  gather_facts: true
  tags: [database-web]

  tasks:
    - name: Install database/web packages
      package:
        name: "[{ item }]"
        state: latest
      with_items:
        - mysql
        - mysql-server
        - MySQL-python
        - php-mysql
        - php-pear
        - php-common
        - php-gd
        - php-devel
        - php
        - php-mbstring
        - php-cli
        - php-process
        - php-snmp
        - net-snmp-utils
        - net-snmp-lib
        - rrdtool

    - name: Start snmpd server
      service:
        name: snmpd
        state: started

    - name: Start mysqld server
      service:
        name: mysqld
        state: started

    - wait_for:
```